

# From Forwarding Tables to Predictive Manifolds: A Routing Analogy for Adapter Warehouses and Self-Models

Patrick Dugan  
(position paper draft)

February 26, 2026

## Abstract

Internet routing protocols traditionally rely on explicit forwarding state: discrete mappings from destinations to next hops that must be recomputed and disseminated under failures and time-varying conditions. Learned and factorized approaches instead represent reachability as compressed predictors, e.g., low-rank embeddings over destination  $\times$  path  $\times$  time. This position paper argues that “adapter warehouses” for large language models (LLMs)—collections of lightweight parameter deltas selected and composed at inference time—instantiate a similar shift from enumerated tables to predictive manifolds. In this view, adapters encode reachability across behavioral or “constitutional” domains in a compressed latent geometry. Removing an adapter resembles node removal in a factorized routing representation: performance degrades via smooth deformation rather than hard hole-punching. We formalize the analogy, identify where it breaks, and propose a research agenda that treats adapter composition as a control-plane problem.

## 1 Motivation

Two pressures push systems away from explicit decision tables and toward compressed predictors:

1. **Scale:** enumerated mappings grow quickly with destinations, policies, and conditions.
2. **Churn:** failures and time variation force frequent updates and cascades.

Internet routing and modular LLM deployment both face these pressures. The architectural question is: *do we store explicit decisions, or do we store a model that can infer them?*

## 2 Routing as tables vs. routing as predictors

### 2.1 Table-based routing (explicit state)

Classical routing can be idealized as a function implemented by explicit state:

$$F(d) \rightarrow \text{next-hop}, \tag{1}$$

computed in the control plane (from policy/topology) and distributed to the data plane. Because state is discrete, failures can induce discontinuities: unreachable destinations, transient loops, and update storms.

### 2.2 Predictive/factorized routing (compressed state)

A predictive approach represents reachability as compressed structure. For example, approximate a destination  $\times$  path  $\times$  time object with low-rank factors:

$$T[d, p, t] \approx \sum_{k=1}^r a_k(d) b_k(p) c_k(t). \tag{2}$$

Representationally, low-rank models turn deletion into deformation: removing a node perturbs an embedding rather than annihilating a table row.

### 3 Adapter warehouses as “fat routing tables”

An adapter warehouse consists of a base model  $M$  and parameter deltas  $\{\Delta_i\}$  (e.g., LoRA updates) selected and composed by a gating/selection mechanism  $g(x)$  given context  $x$ . A minimal abstraction is:

$$y = M(x) + \sum_i g_i(x) \Delta_i. \quad (3)$$

(Static LoRA corresponds to constant  $\Delta_i$ ; more general adapters may be conditional.)

#### 3.1 Reachability across behavioral / constitutional domains

In deployment, adapters often correspond to regimes such as style/persona, task specialization, instruction-following, and safety/constitutional constraints (“what outputs are reachable from which prompts”). The warehouse therefore stores a *basis* of behavioral moves in weight space, reweighted as a function of  $x$ , rather than an explicit lookup for every context.

#### 3.2 Analogy summary

Table 1 maps networking concepts to modular LLMs.

Table 1: Routing analogy for adapter warehouses.

| Networking              | Modular LLMs (adapters/self-models)   |
|-------------------------|---------------------------------------|
| Destination $d$         | Prompt/task context $x$               |
| Next-hop decision       | Adapter selection/composition         |
| Control-plane state     | Adapter inventory + gating policy     |
| Routing table explosion | Adapter sprawl + selection complexity |
| Node/link removal       | Missing adapter / disabled capability |
| Hole-punching failure   | Abrupt capability cliffs              |
| Low-rank embedding      | Low-dimensional basis in weight space |
| Graceful degradation    | Smooth fallback via reweighting       |

**Claim.** An adapter warehouse behaves more like a compressed predictive model of behavioral reachability than an enumerated table of behaviors.

### 4 Failure semantics: hole-punching vs. deformation

In explicit table systems, removing an entry can create a hard failure. Analogously, if capability depends on a single monolithic fine-tune or brittle mode switch, removal can cause abrupt collapse. In a factorized representation, removal is closer to deleting one component of a basis: remaining components can partially approximate the target behavior, yielding graded failure. For adapters, disabling one component ideally shifts mixture weights toward nearby directions; in practice, interference and miscalibrated gating can still create discontinuities, so graceful degradation is a design goal rather than a guarantee.

### 5 Where the analogy breaks (usefully)

The analogy fails in ways that point to missing theory:

1. **Physics vs. approximation:** networks obey hard constraints; behavior manifolds can fold and collide, so “nearby” deltas may not be substitutable.
2. **Convergence:** routing demands stability/loop-freedom; adapter gating can flap or oscillate without formal guarantees.
3. **Observability:** network metrics are direct; behavioral distance requires probing and evaluation instrumentation.

## 6 Research agenda: treat adapter composition as a control plane

Viewing adapter composition as routing suggests concrete questions.

### 6.1 Behavioral routing metrics

Define analogues of routing metrics:

- **Reachability:** for constraints  $C$ , what fraction of contexts  $x$  yield acceptable outputs?
- **Stretch:** extra tokens/latency to reach correctness under constraints.
- **Congestion:** overuse of certain adapter directions that induces interference or loss of generality.

### 6.2 Stability criteria for gating

Develop constraints on  $g(x)$  to prevent:

- oscillatory switching across turns (policy flapping),
- unstable mixtures that amplify errors under distribution shift,
- cascades where small prompt changes trigger large behavior jumps.

### 6.3 Topology-aware adapter geometry

Construct embeddings where distance corresponds to substitutability:

- local neighborhoods support fallback,
- removal yields bounded degradation,
- composition obeys smoothness (e.g., regularization/orthogonalization).

### 6.4 Control-plane / data-plane separation

Separate (and version/audit) the *control plane* (selection rules, gating, evaluation probes) from the *data plane* (base inference plus applied deltas), mirroring routing practice.

## 7 Conclusion

Adapter warehouses and self-model architectures can be framed as a representational shift from explicit decision tables to compressed predictors over latent geometry. The routing analogy foregrounds graceful degradation under component loss and suggests importing control-plane thinking: stability constraints for gating, behavioral reachability metrics, and topology-aware embeddings that make robustness a property of representation.